How the NBA Gradually Learned to Pay for Three-Point Skill:

Evidence from Era Evolution and Positional Heterogeneity

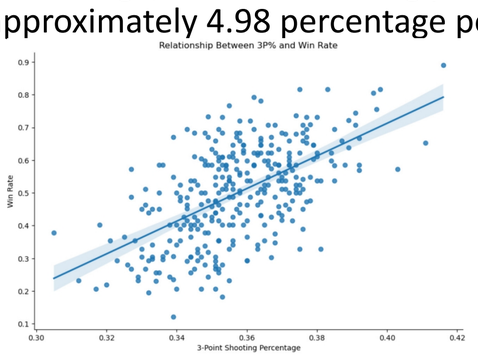
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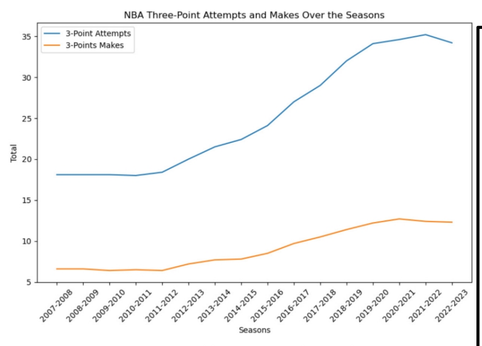
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1.Background

Over the past two decades, the NBA has undergone a major strategic transformation. Three-point shooting has evolved from a complementary scoring option into a central source of spacing, offensive efficiency, and roster value. According to Hu (2024), the number of three-point attempts and made in the NBA has surged in recent years, and the three-point shooting percentage plays a crucial role in a team's success. For every 1% increase in three-point shooting percentage, a team's winning percentage is expected to increase by approximately 4.98 percentage points.





Yet changes in how the game is played do not automatically imply changes in how the labor market rewards player skills. This poster asks not simply who shoots better, but when the NBA began to systematically pay for three-point skill, and whether that process unfolded differently across positions.

2. Research Question

This poster addresses two main questions. First, did the NBA labor market gradually develop an independent salary premium for three-point skill across different historical eras? Second, did that premium emerge in the same way for Guards, Forwards, and Centers, or did it vary by position? The broader goal is to understand how market pricing of three-point skill evolved over time, and how that evolution differed across roles on the court.

3. Data and Sample

The analysis uses NBA player-season performance and salary data from 2000 to 2025. To reduce noise from extremely low-end contracts and fringe roster spots, the sample is restricted to player-seasons where nominal salary is at least 1.5% of the league salary cap in that year. Players are grouped into three positional categories—Guard, Forward, and Center—and the sample is divided into four historical periods:

2000–2010；2011–2015；2016–2020；2021–2025

This structure allows the analysis to trace both era-based change and position-specific differences.

4. Empirical Framework

The dependent variable is the log of normalized salary, defined as player salary relative to the same-year league average salary. Three-point skill and free-throw skill are measured using composite indices constructed as z-scored makes plus z-scored percentage, each averaged over a three-year rolling window. Two-point ability is not treated as a single composite measure; instead, it is separated into two-point makes and two-point percentage in order to distinguish market pricing of volume from pricing of efficiency.

The model controls for rebounds, assists, steals, blocks, age, and age squared, and includes year fixed effects and team fixed effects. Standard errors are clustered at the player level. In the final specification, the model also includes a team-level budget variable:

$$\text{Team Salary Pressure} = \frac{\text{Team Payroll}}{\text{Salary Cap}}$$

This variable captures the extent to which a team is operating under a tighter payroll environment.

Regression Model:

$$ln(\frac{Salary_{ijt}}{Salary_t}) = \alpha + \beta_1 Skill3P_{ijt}^{(3)} + \beta_2 Z(2Pmade)_{ijt}^{(3)} + \beta_3 Z(2P\%)_{ijt}^{(3)} + \beta_4 SkillFT_{ijt}^{(3)} + \beta_5 TRB_{ijt}^{(3)}$$

$$+ \beta_6 AST_{ijt}^{(3)} + \beta_7 STL_{ijt}^{(3)} + \beta_8 BLK_{ijt}^{(3)} + \beta_9 AgeC_{ijt}^{(3)} + \beta_{10}(AgeC_{ijt}^{(3)})^2 + \beta_{11} TeamPressure_{jt}^{(3)}$$

$$+ \beta_{12} NearMax25_{ijt} + \delta_t + \gamma_j + \varepsilon_{ijt}$$

The sample is restricted to player-season observations with salary at least 1.5% of the same-year NBA salary cap.

$$CapShare_{ijt} = \frac{Salary_{ijt}}{SalaryCap_t} \geq 0.015 \quad NearMax25_{ijt} = 1(\frac{Salary_{ijt}}{SalaryCap_t} \geq 0.25)$$

Our model also includes a near-max salary proxy, which equals one when salary reaches at least 25% of the same-year salary cap.

5. Main Historical Pattern

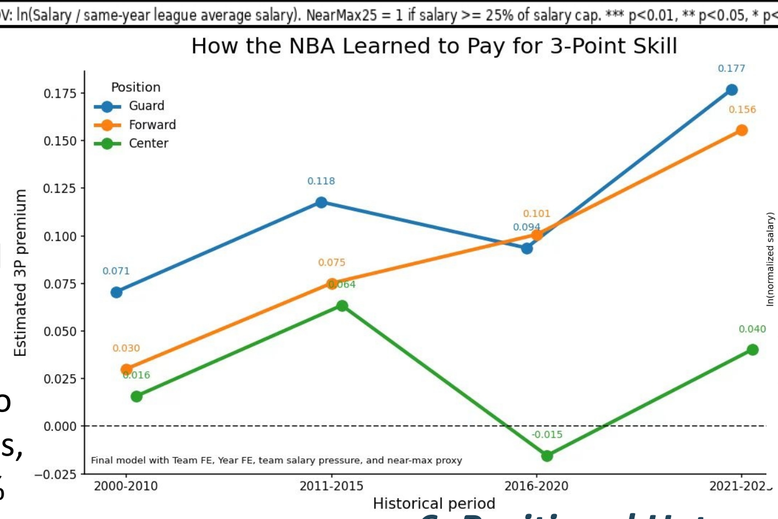
The central finding is that the NBA did not always pay for three-point skill in the way it does today. Instead, the salary premium for three-point ability emerged gradually over time.

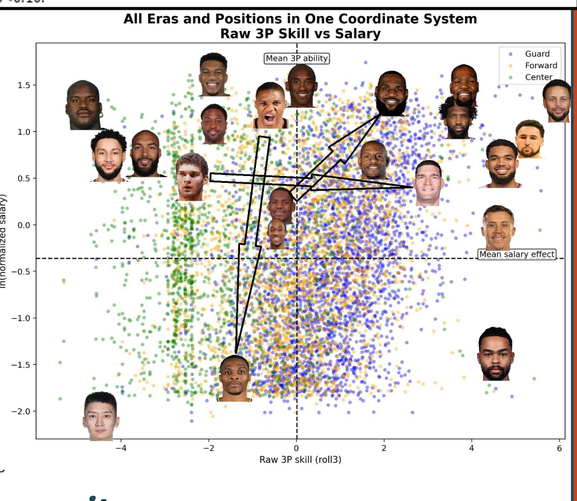
In 2000–2010, the premium is mainly visible among Guards, suggesting that during this period three-point shooting was still priced largely as part of the guard skill set rather than as a universally valued asset across the league. In 2011–2015, the premium begins to expand beyond guards and becomes more visible among forwards, indicating that the market was starting to place greater value on spacing-oriented players. In 2016–2020 and especially 2021–2025, the pricing of three-point skill becomes much more established, particularly for guards and forwards.

From a historical perspective, then, the market valuation of three-point skill was not constant; it was built over time.

	Guard				Forward				Center			
Variable	00-10	11-15	16-20	21-25	00-10	11-15	16-20	21-25	00-10	11-15	16-20	21-25
3P composite index	0.071***	0.118***	0.094***	0.177***	0.030*	0.075***	0.101***	0.156***	0.016	0.064*	-0.015	0.040
2P makes (z)	0.319***	0.326***	0.350***	0.331***	0.076	0.306***	0.325***	0.192***	0.172**	0.039	0.140*	0.228*
2P% (z)	-0.048	0.006	-0.100**	-0.102***	-0.035	-0.068*	-0.139***	-0.005	-0.011	-0.102	0.001	0.076
FT composite index	-0.012	0.030	0.015	-0.054	0.041	0.030	0.004	0.057*	0.023	0.075**	0.033	0.006
Rebounds	0.036	0.056	0.001	0.036	0.096***	0.047**	0.047*	0.049**	0.073**	0.147***	0.099***	0.068**
Assists	0.082***	0.052***	-0.003	0.060***	0.083***	0.107**	0.050	0.093**	-0.038	0.019	0.017	0.087
Steals	-0.077	0.077	0.348***	0.307***	-0.057	-0.088	0.241**	0.229**	0.239	0.185	-0.006	0.170
Blocks	0.349**	0.236	0.368	0.275*	0.062	0.240***	0.053	0.238*	0.132*	0.205**	0.181	0.207*
Centered age	0.078***	0.052***	0.054***	0.049***	0.075***	0.050***	0.070***	0.043***	0.058***	0.069***	0.073***	0.051***
Centered age squared	-0.005***	-0.003**	-0.005***	-0.002	-0.007***	-0.005***	-0.006***	-0.002*	-0.008***	-0.015***	-0.008***	-0.004**
Team salary pressure	0.091	0.174	0.377*	0.105	-0.178*	0.400*	0.117	0.195	0.088	-0.017	0.860***	0.137
Near-max proxy	0.661***	0.703***	0.609***	0.694***	0.772***	0.464***	0.473***	0.579***	0.736***	0.442***	0.618***	0.760***
Observations	1077	634	621	755	1145	647	600	692	491	267	301	349
R-squared	0.680	0.683	0.608	0.711	0.653	0.641	0.655	0.718	0.643	0.751	0.663	0.685
Adj. R-squared	0.662	0.657	0.577	0.693	0.636	0.613	0.627	0.698	0.598	0.697	0.604	0.638

DV: ln(Salary / same-year league average salary). NearMax25 = 1 if salary >= 25% of salary cap. *** p<0.01, ** p<0.05, * p<0.10.





6. Positional Heterogeneity

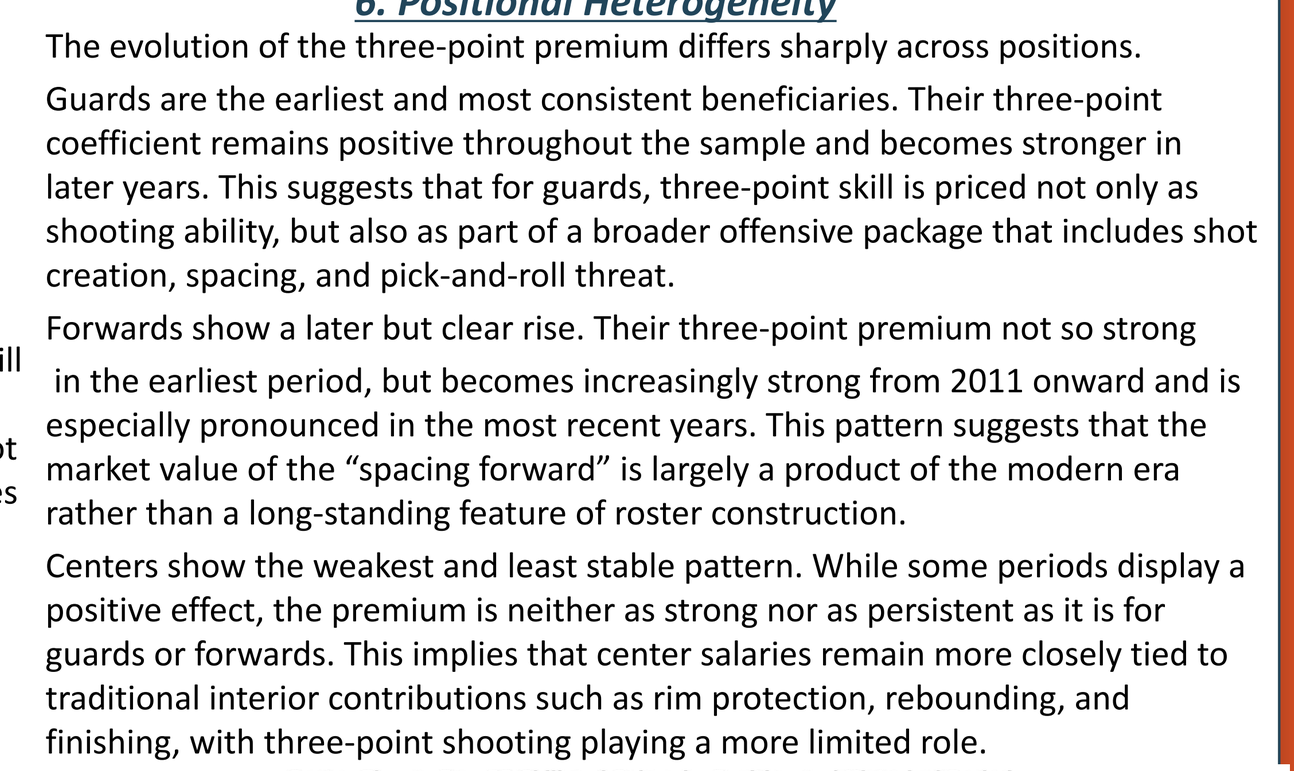
The evolution of the three-point premium differs sharply across positions.

Guards are the earliest and most consistent beneficiaries. Their three-point coefficient remains positive throughout the sample and becomes stronger in later years. This suggests that for guards, three-point skill is priced not only as shooting ability, but also as part of a broader offensive package that includes shot creation, spacing, and pick-and-roll threat.

Forwards show a later but clear rise. Their three-point premium not so strong in the earliest period, but becomes increasingly strong from 2011 onward and is especially pronounced in the most recent years. This pattern suggests that the market value of the “spacing forward” is largely a product of the modern era rather than a long-standing feature of roster construction.

Centers show the weakest and least stable pattern. While some periods display a positive effect, the premium is neither as strong nor as persistent as it is for guards or forwards. This implies that center salaries remain more closely tied to traditional interior contributions such as rim protection, rebounding, and finishing, with three-point shooting playing a more limited role.

Poster Figure: Raw 3P Skill and Salary by Position and Historical Period (with fitted regression lines)



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